Lecture 19 (10/30)

Interactions and residual plots

The course so far

So far:

- **Unit 1:** terminology and notation (probability, random variables, conditioning).
- **Unit 2:** linear regression as a model of conditional probabilities,

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2).$$

- **Unit 3:** how to fit the model and quantify uncertainty (estimators, standard errors, tests and confidence intervals).
- **Unit 4:** Multiple predictor linear regression,

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Next:

- **Unit 5:** nonlinear models and overfitting. Machine learning vs. statistics.
- **Unit 6:** Bayesian statistics.

Today:

1) Interactions (test scores, mother's schooling and IQ)
2) Residual plots to identify interactions and other model deficiencies

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- The association between earnings and height might depend on gender.
- The effect of a drug for an infection might depend on whether someone has had the infection before.

These are interactions and are not captured by our existing models.

Example

Test scores, revisited. Survey of 434 children (from *Regression and Other Stories*):

- Y (`kid_score`): the child's test score;
- X_{hs} (`mom_hs`): 1 if the mother attended high school, 0 if not;
- X_{iq} (`mom_iq`): the mother's IQ.

Without interactions, the model output was

Dep. Variable:	kid_score	R-squared:			0.214
	coef	std err	t	P>|t|	[0.025	0.975]
const	25.7315	5.875	4.380	0.000	14.184	37.279
mom_hs	5.9501	2.212	2.690	0.007	1.603	10.297
mom_iq	0.5639	0.061	9.309	0.000	0.445	0.683

Interaction terms

Let the slope vs. X_{iq} depend linearly on X_{hs}:

$$\begin{aligned} Y &= \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + (\beta_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}}) X_{\text{iq}} + \epsilon \\ &= \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} X_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}} X_{\text{iq}} + \epsilon. \end{aligned}$$

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Question

Compute

$$E[Y \mid X_{\mathrm{hs}}{=}1, X_{\mathrm{iq}} = x] - E[Y \mid X_{\mathrm{hs}}{=}0, X_{\mathrm{iq}} = x].$$

Fitting the interaction model

Dep. Variable:		kid_score	R-squared:			0.230
	coef	std err	t	P>|t|	[0.025	0.975]
const	-11.4820	13.758	-0.835	0.404	-38.523	15.559
mom_hs	51.2682	15.338	3.343	0.001	21.122	81.414
mom_iq	0.9689	0.148	6.531	0.000	0.677	1.260
hs_iq	-0.4843	0.162	-2.985	0.003	-0.803	-0.165

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Question

- What is the slope of score vs. mother's IQ in each schooling group?
- How does the effect of high school depend on IQ?
- Why is $\hat{\beta}_{\text{hs}} \approx 51$ so different from $\hat{\beta}_{\text{hs}} \approx 5.95$ in the model without the interaction?

Centering a predictor

Replace X_{iq} by the centered predictor $\Delta_{\text{iq}} = X_{\text{iq}} - \overline{X}_{\text{iq}}$ (here $\overline{X}_{\text{iq}} = 100$) and fit

$$Y = \beta_0 + \beta_{\text{hs}} X_{\text{hs}} + \beta_{\text{iq}} \Delta_{\text{iq}} + J_{\text{hs,iq}} X_{\text{hs}} \Delta_{\text{iq}} + \epsilon.$$

Dep. Variable:		kid_score	R-squared:			0.230
	coef	std err	t	P>|t|	[0.025	0.975]
const	85.4069	2.218	38.502	0.000	81.047	89.767
mom_hs	2.8408	2.427	1.171	0.242	-1.929	7.610
iq_c	0.9689	0.148	6.531	0.000	0.677	1.260
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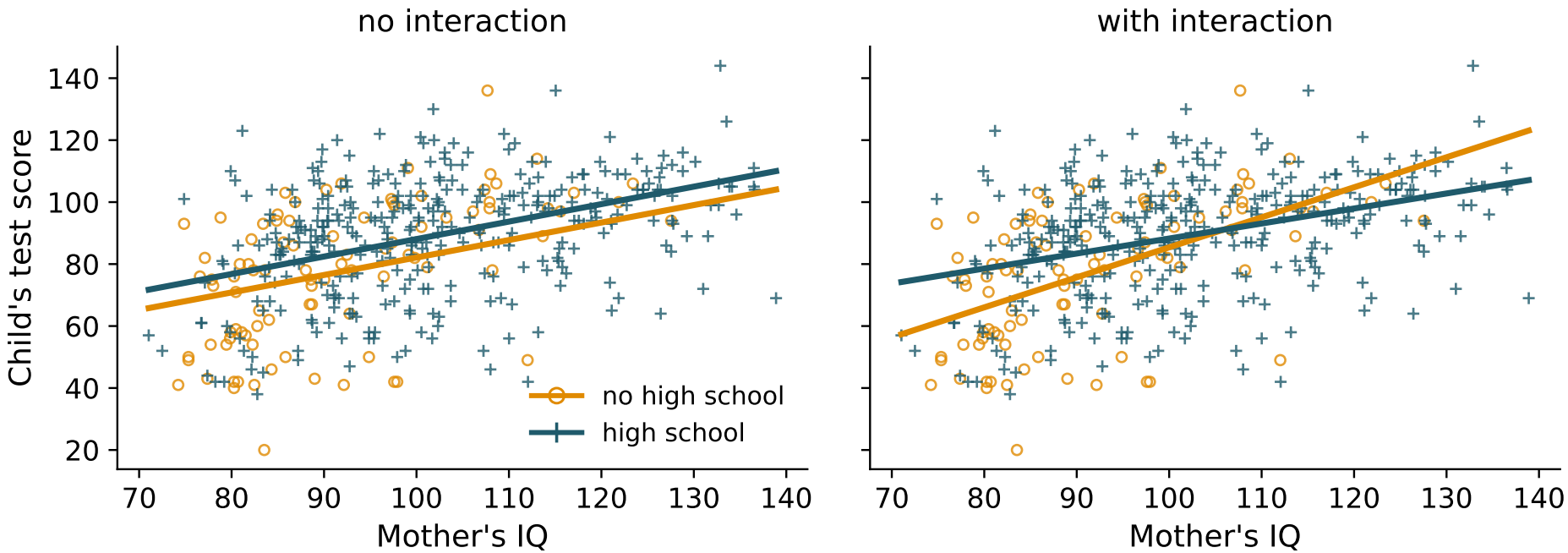
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Question

Interpret $\hat{\beta}_0$ and $\hat{\beta}_{\text{hs}}$ in the output above.

Visualizing regression lines with interaction



Without the interaction the fitted lines are parallel.

Example: earnings, height and gender

Survey of 1629 adults with positive earnings (from *Regression and Other Stories*): Y is the log of earnings in \$1000s, X_{height} is height in inches, and $X_{\text{male}} = 1$ for men, 0 for women. Without an interaction:

Dep. Variable:		log_earnings	R-squared:			0.080
	coef	std err	t	P>|t|	[0.025	0.975]
const	1.0686	0.506	2.111	0.035	0.076	2.062
height	0.0238	0.008	3.044	0.002	0.008	0.039
male	0.3715	0.062	6.023	0.000	0.251	0.493

With an interaction, and height centered at its mean (66.8 inches):

Dep. Variable:		log_earnings	R-squared:			0.081
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.6405	0.037	71.856	0.000	2.568	2.713
height_c	0.0148	0.011	1.389	0.165	-0.006	0.036
male	0.3568	0.063	5.684	0.000	0.234	0.480
male_height_c	0.0196	0.016	1.251	0.211	-0.011	0.050

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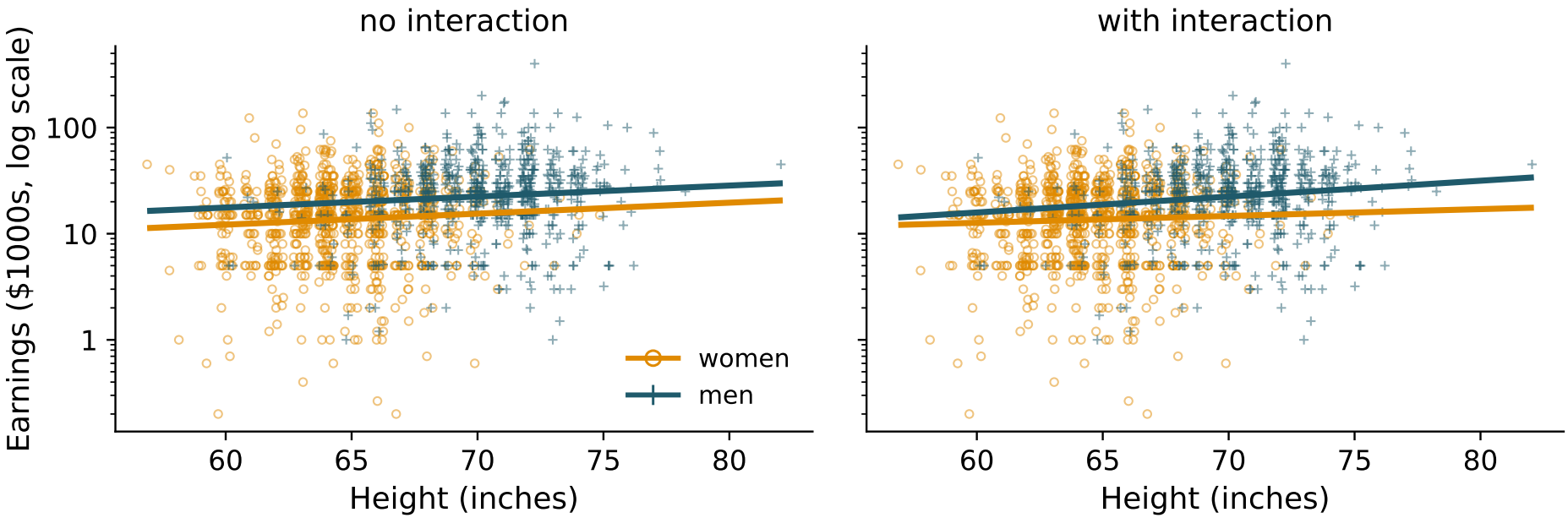
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Question

What is the slope of log earnings vs. height for women? For men? What is the difference between men and women of average height? Interpret each as a percent change in earnings.

Earnings: no interaction vs. interaction



Residual plots

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The residuals are what the fitted model leaves unexplained:

$$r_j = Y_j - \hat{y}_j, \qquad \hat{y}_j = \hat{\beta}_0 + \sum_{k=1}^{K} \hat{\beta}_k X_{k,j}.$$

(r_j is computed from the data; ϵ_j in the model cannot be observed.)

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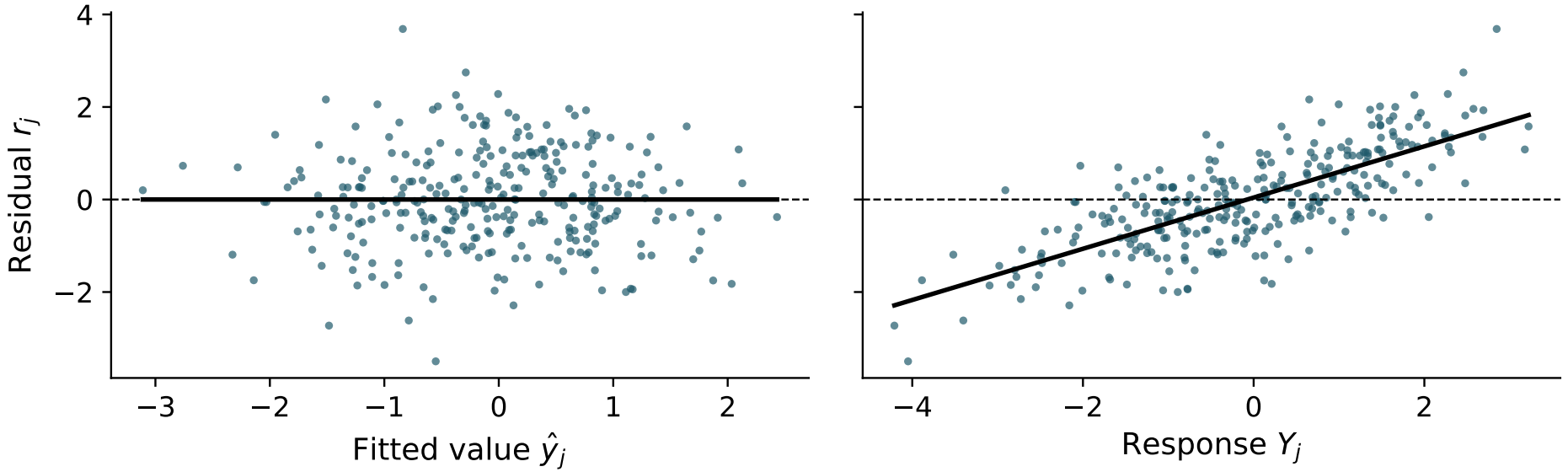
(r_j is computed from the data; ϵ_j in the model cannot be observed.)

If the model is right, r_j is approximately Normal$(0, \sigma_\epsilon^2)$ *whatever the values of the predictors*. A residual plot is meant to visualize the r_j.

Residuals vs. fitted values, not vs. Y

Consider

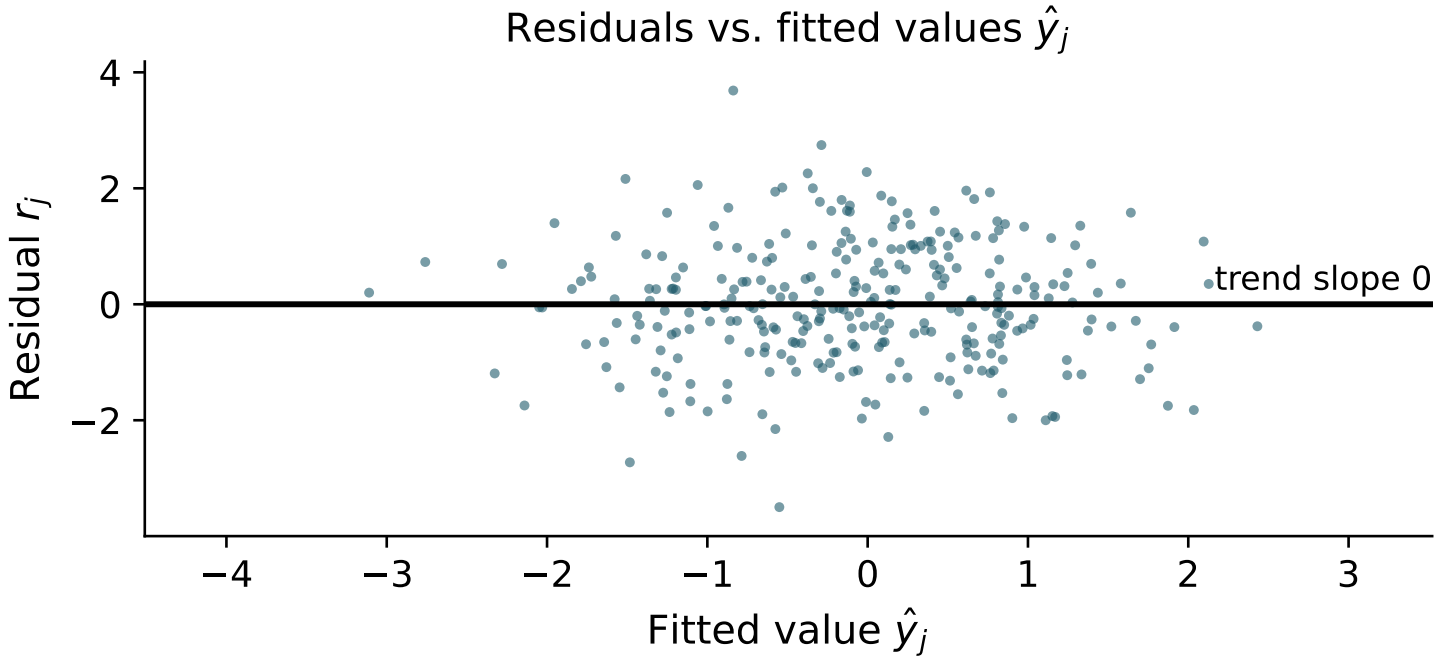
$$Y = 0.2X_1 + 20X_2 + \epsilon, \quad \sigma_\epsilon = 1$$



Question

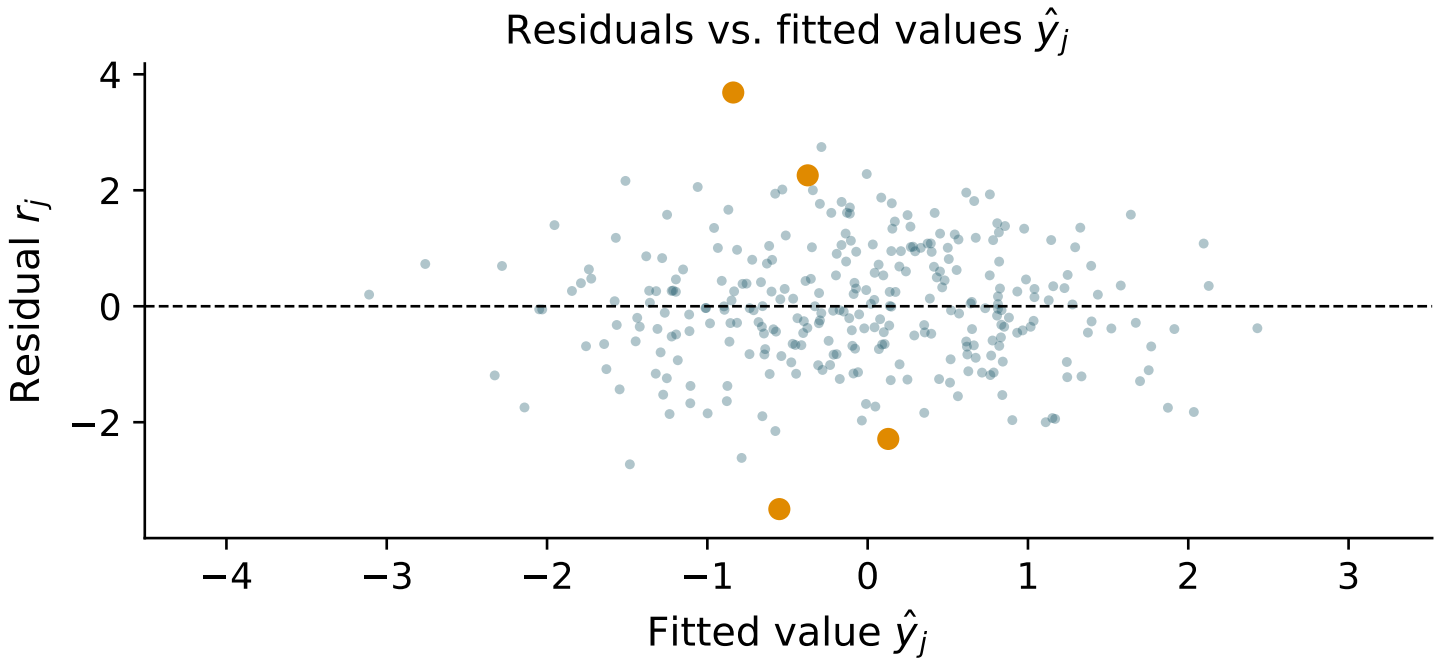
Why is there a trend on the right plot?

Why plot $\hat{y}_i$ on x axis



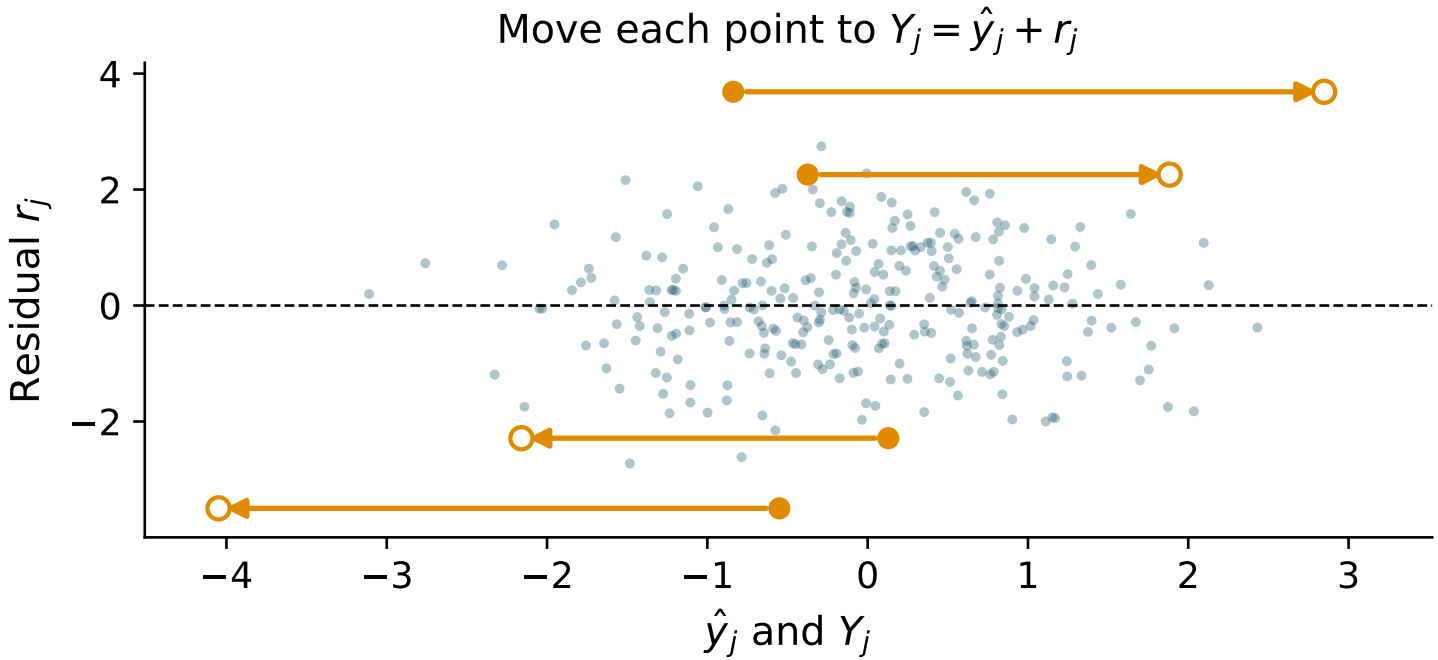
1. If the model is right, r_j has the same distribution whatever the predictors are, so vs. $\hat{y}_j$ we expect a symmetric spread in the vertical direction.

Why plot $\hat{y}_i$ on x axis



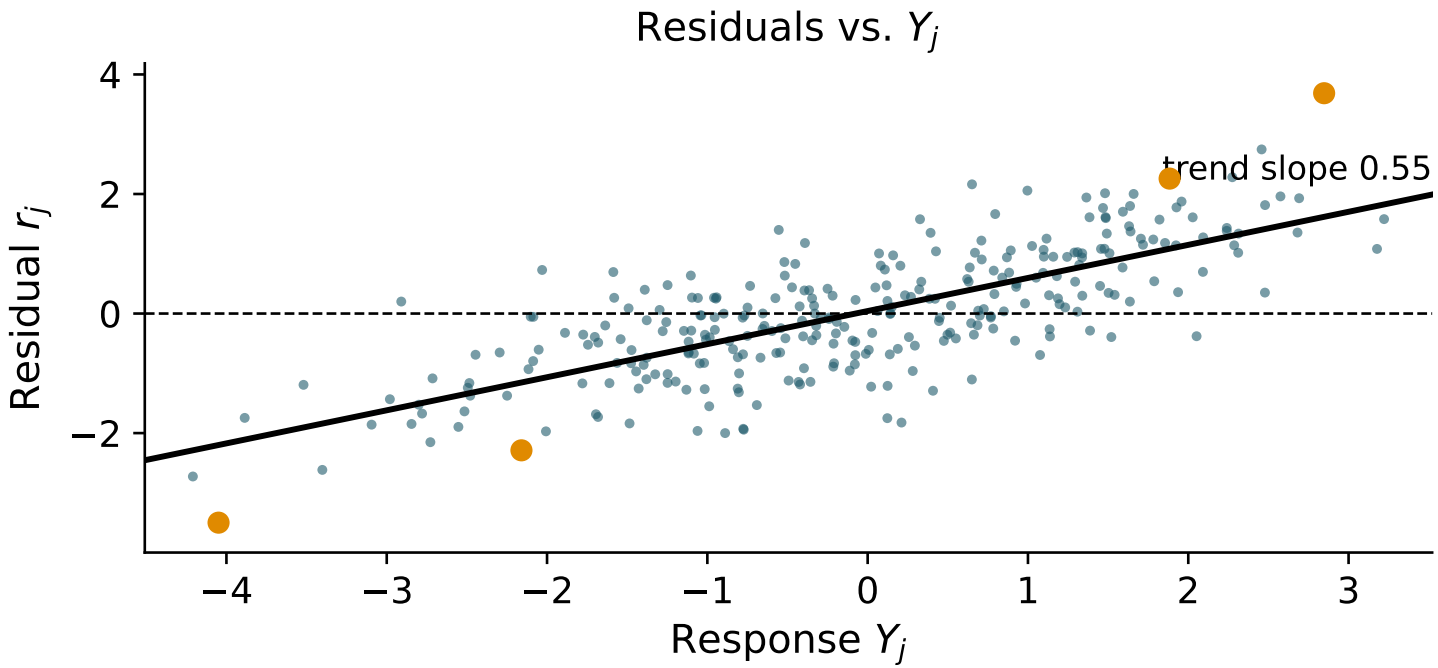
2. Pick out a few points with large residuals.

Why plot $\hat{y}_i$ on x axis



3. Since $Y_j = \hat{y}_j + r_j$, putting Y_j on the horizontal axis shifts each point sideways by the corresponding residual.

Why plot $\hat{y}_i$ on x axis



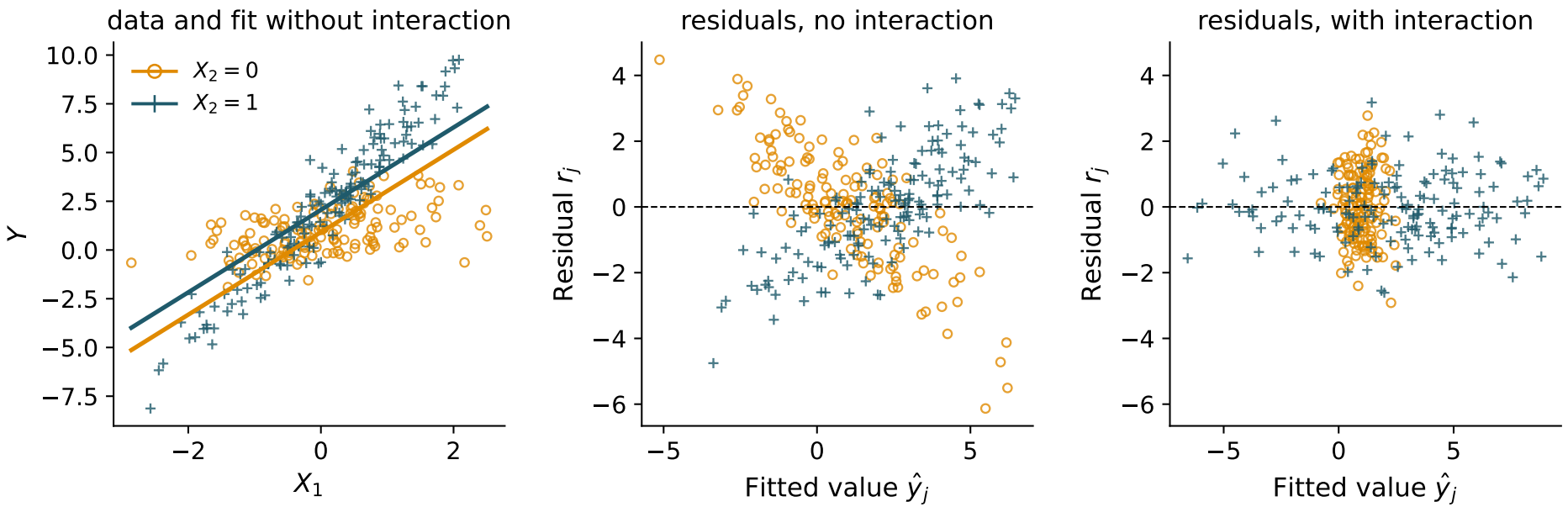
4. Every point shifts this way, so the cloud tilts even though the model is correct.

Example: binary predictor with an interaction

Consider

$$Y = 1 + 0.5X_1 + X_2 + 3X_1X_2 + \epsilon$$

whre X_2 is a binary predictor. Below is the data fit without an interaction and the residual plots with and without.

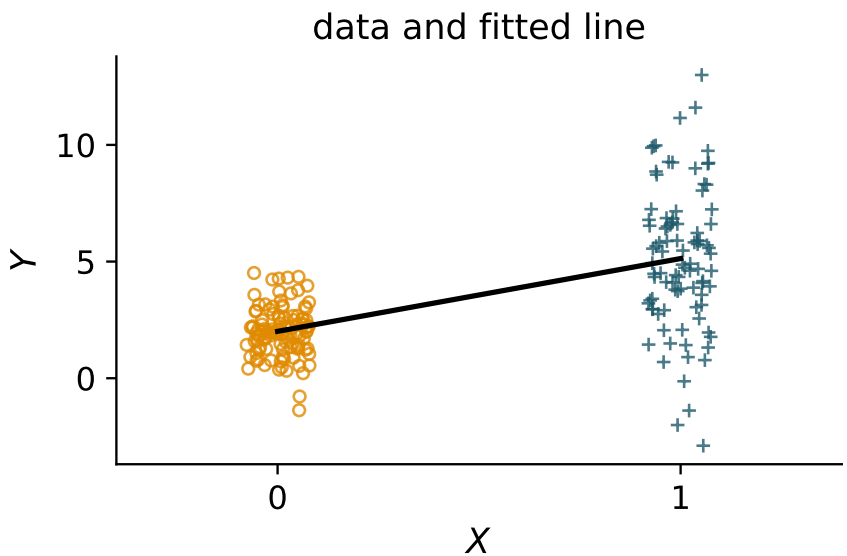


Example: binary predictor with unequal variances

Consider

$$Y = 2 + 3X + \epsilon(1 + 2X)$$

Below is the data with the fitted line and the residual plot. With one binary predictor, $\hat{y}_j$ takes only two values.

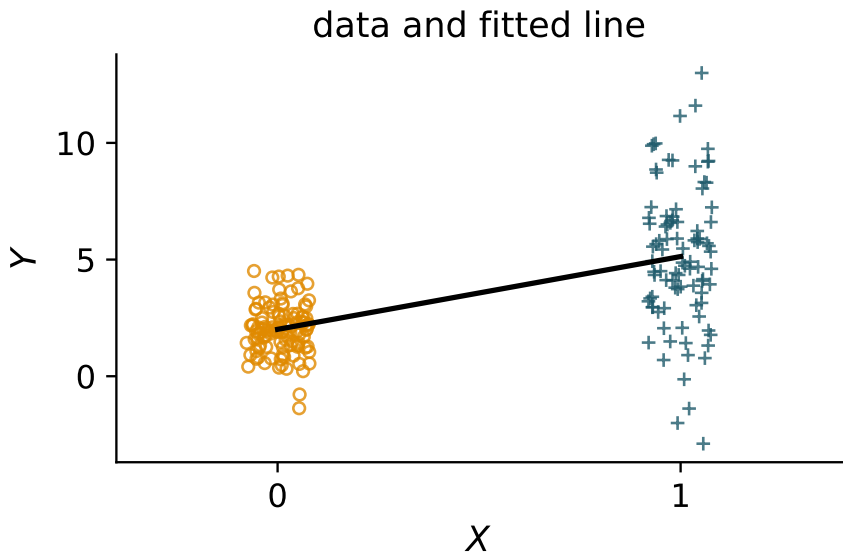


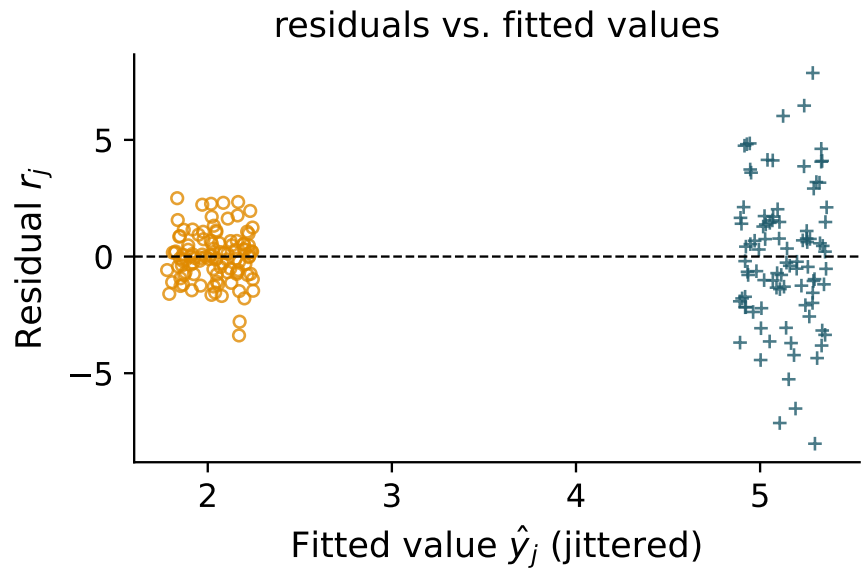
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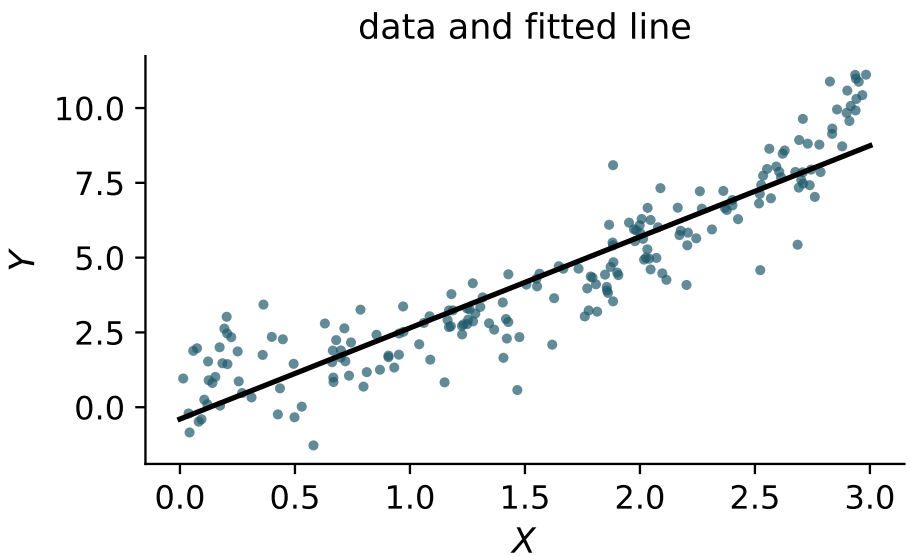


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$$Y = 1 + X^2 + \epsilon$$

where X is between 0 and 3, fitted with a straight line. Below is the data with the fitted line and the residual plot.

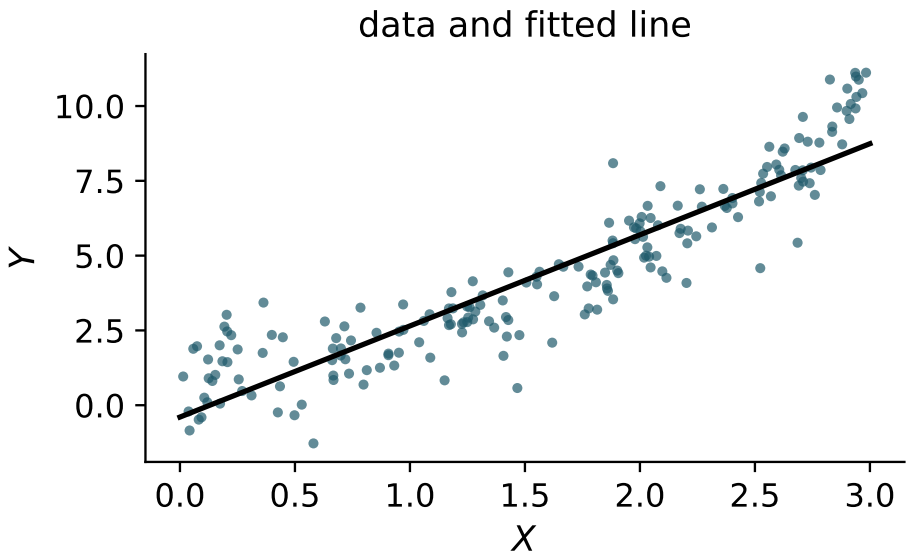


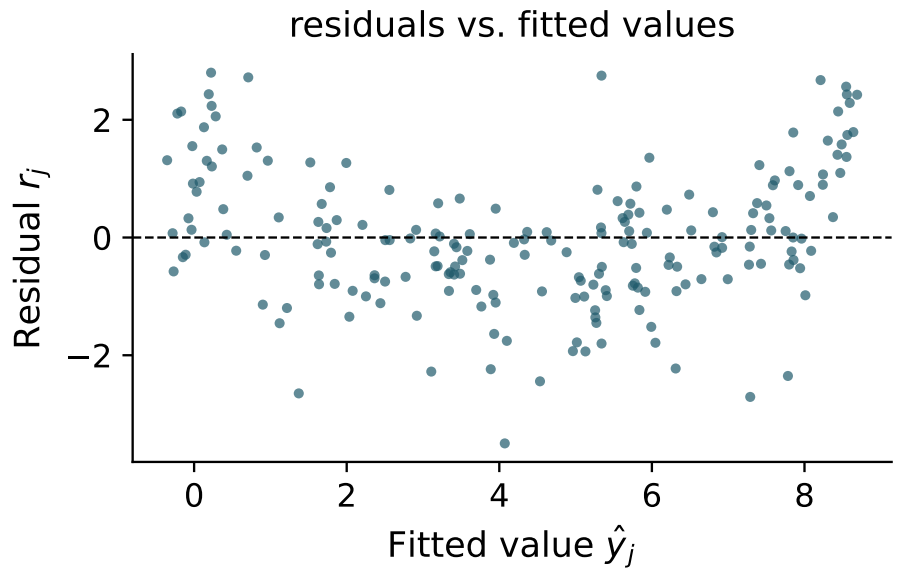
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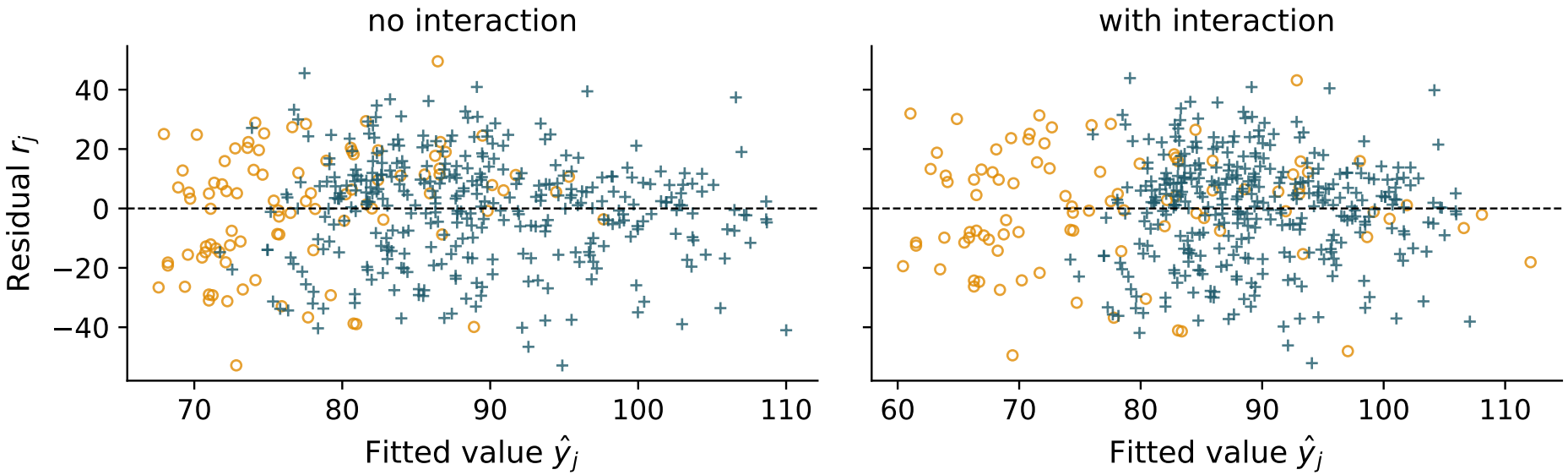
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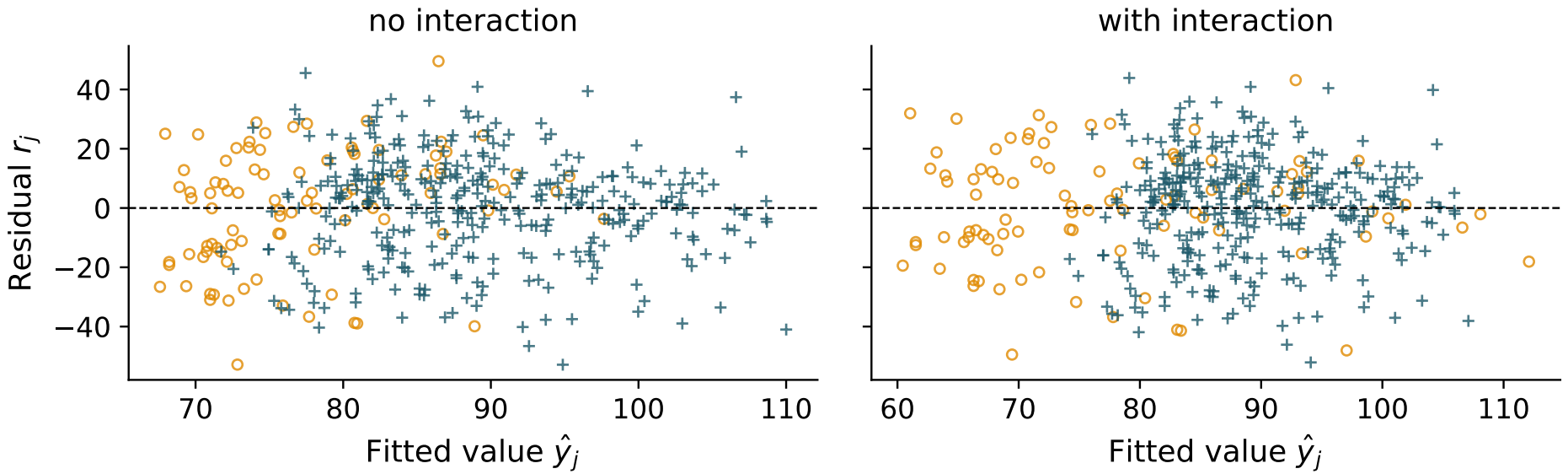




Test scores: residuals vs. fitted values



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Question

The missing interaction is not evident from the residual plot, but it is statistically significant ($p = 0.003$). What does this suggest about how R^2 changes when adding the residual?

Example: advertising budgets and sales

Sales of a product in 200 markets (from *An Introduction to Statistical Learning*): Y is sales in thousands of units, and X_{TV}, X_{radio} are the TV and radio advertising budgets in \$1000s. Without an interaction:

Dep. Variable:		sales	R-squared:			0.897
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.9211	0.294	9.919	0.000	2.340	3.502
TV	0.0458	0.001	32.909	0.000	0.043	0.048
radio	0.1880	0.008	23.382	0.000	0.172	0.204

With an interaction, and both budgets centered at their means (147.0 and 23.3):

Dep. Variable:		sales	R-squared:			0.968
	coef	std err	t	P>|t|	[0.025	0.975]
const	13.9470	0.067	208.737	0.000	13.815	14.079
TV_c	0.0444	0.001	56.673	0.000	0.043	0.046
radio_c	0.1886	0.005	41.806	0.000	0.180	0.198
TV_radio_c	0.0011	5.24e-05	20.727	0.000	0.001	0.001

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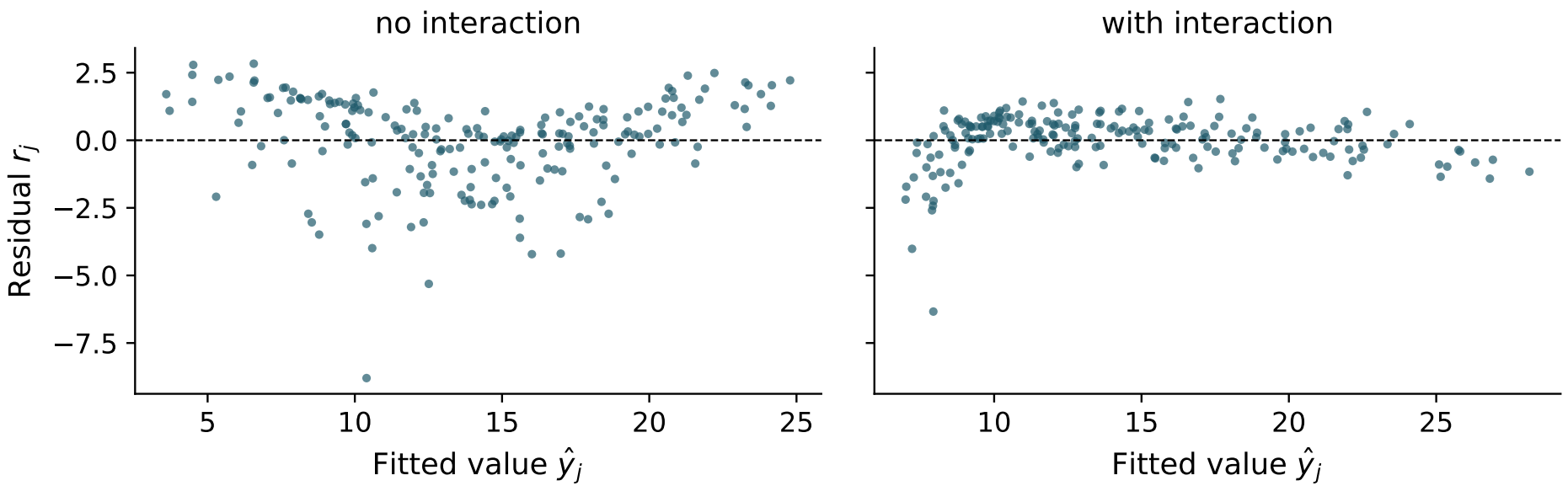
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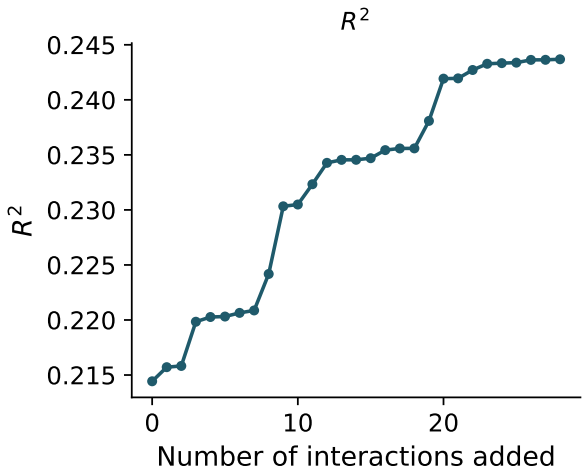
How does the effect of an extra \$1000 of radio advertising on sales depend on the TV budget?

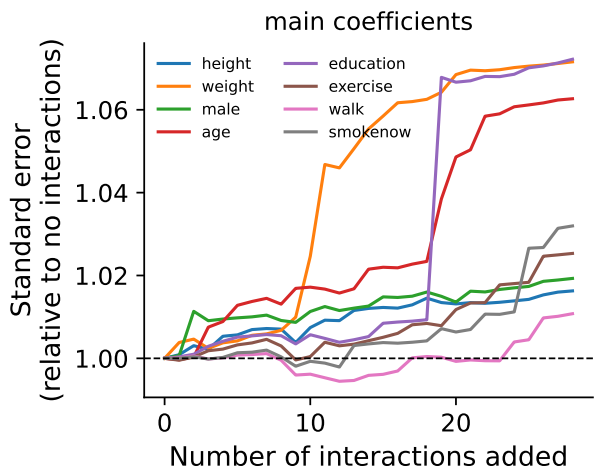
Advertising: residuals without vs. with interaction

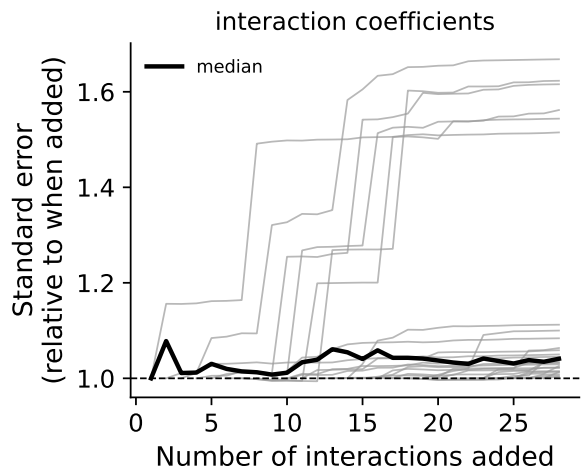


Earnings: adding every interaction

Consider log earnings with eight predictors: height, weight, gender, age, education, exercise, walking and smoking. There are 28 possible pairwise interactions. Below, they are added one at a time, using all 1604 people with complete data.

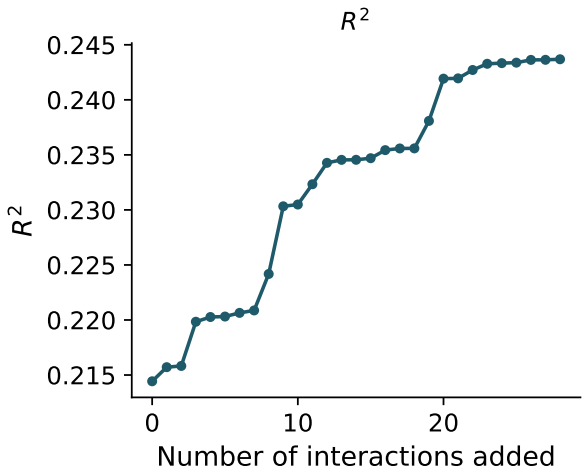


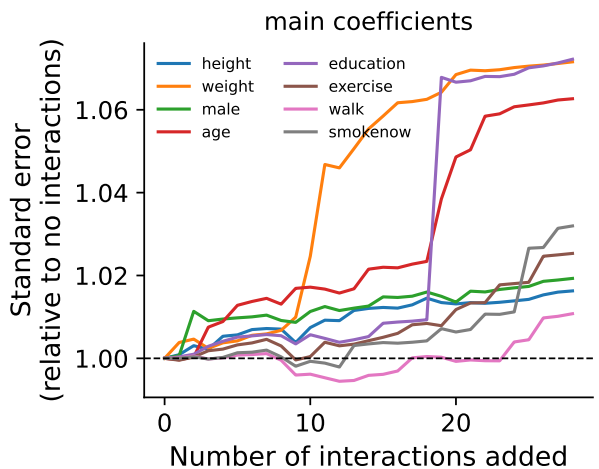


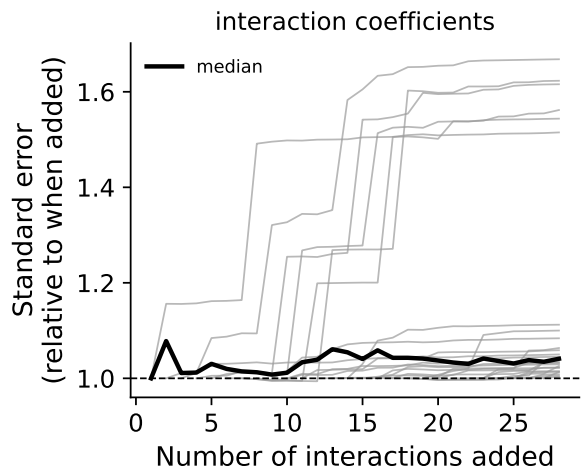


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Question

Is the model with all 28 interactions better than the one with none? What happens to the uncertainty in each coefficient?